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title: "Why Mullvad VPN" slug: "why-mullvad-vpn"

Chapter 1 — Why Mullvad VPN

Learning Objectives

By the end of this chapter, you'll be able to:

- Describe Mullvad VPN's mission, ownership, and history in your own words
 - Articulate what makes Mullvad different from other commercial VPNs
 - Match Mullvad's features to your personal privacy threat model
 - Decide whether Mullvad is the right VPN for your use case
-

Part A — Detailed Information

A.1 What Mullvad VPN Is

Mullvad VPN is a commercial VPN service operated by **Mullvad VPN AB**, a Swedish company registered in Gothenburg under corporate number **559238-4001**. Founded in **March 2009**, Mullvad was among the first consumer VPNs to be designed around an account-number-only login model — no email, no password, no real name.

The company is wholly owned by **Amagicom AB**, which is in turn owned by its two founders: **Fredrik Strömberg** and **Daniel Berntsson**. Both founders remain actively involved in the company. The parent company's name, "Amagicom," comes from the Sumerian word *ama-gi*, which is the oldest recorded word for "freedom" — fitting for a company whose central mission is to defend personal privacy.

The current Mullvad website is at <https://mullvad.net>



, and they also operate a Tor onion

service for users who want to visit the site anonymously.

A.2 Why It Matters (Threat Model Fit)

Mullvad's design choices map directly onto the threat models of:

- **Casual users** who don't want their ISP to log their browsing history
- **Journalists** who need to protect sources and themselves
- **Activists** who face state-level surveillance
- **Anyone in a censorship-heavy country** who needs reliable internet access
- **Privacy advocates** who believe data minimization is a fundamental right

What sets Mullvad apart from competitors is a combination of choices, not a single killer feature:

1. **Anonymous payment** — cash in envelope, Monero, Bitcoin
2. **No-logs policy** — independently audited by Cure53 (2020) and Assured AB (2023)
3. **Account-number-only login** — no email, no password, no recovery option
4. **RAM-only infrastructure** — servers cannot retain data after reboot
5. **Flat €5/month pricing** — no upsells, no "premium" tier
6. **Open source** — both client apps and significant server-side code
7. **Swedish jurisdiction + 14 Eyes** — controversial but combined with no logs, moot

"Mullvad VPN exists to fight against mass surveillance and censorship. We do this primarily through our VPN service and our browser. We believe in a free and open society where people have the right to privacy; the right to their own beliefs and thoughts, and the right to decide for themselves exactly when and with whom they want to share them." — Mullvad Press page

A.3 Architecture (Brief Overview)

Mullvad's network consists of two distinct populations of servers:

- **553 WireGuard servers** (live as of 2026-09-28) — the primary network. Each server has a unique hostname in the form `xx-xyz-wg-###.relays.mullvad.net` (e.g., `se-mma-wg-001`, `nl-ams-wg-001`). Each exposes two SOCKS5 proxies (one local-only at `10.64.0.1:1080` and one peer-reachable at `10.124.0.x:1080`) and one WireGuard tunnel endpoint at a unique UDP port (`multihop_port`).
- **13 Bridge servers** — OpenVPN + Shadowsocks endpoints designed to bypass censorship. Naming: `xx-xyz-br-###.relays.mullvad.net`.

You'll learn the full architecture in detail in Part IV (WireGuard + SOCKS5 Internals) of this book, but for now the key facts are:

Server type	Count	Tunnel protocol	SOCKS5 endpoint
WireGuard	553	WireGuard (UDP, port varies)	<code>10.64.0.1:1080</code> + <code>10.124.0.x:1080</code>
Bridge	13	OpenVPN + Shadowsocks	None

A.4 Specifications

Property	Value
Company	Mullvad VPN AB
Registration	559238-4001 (Sweden)
Headquarters	Box 53049, 400 14 Gothenburg, Sweden
Parent company	Amagicom AB
Founded	March 2009
Pricing	€5/month flat (no upsells)
Server count	553 WireGuard + 13 Bridges
Protocols	WireGuard (preferred), OpenVPN (Bridge mode only)
Account model	16-digit account number (no email, no password)
Payment methods	Cash, Monero, Bitcoin, voucher, credit card
Audit history	Cure53 (2020), Assured AB (2023)
Open source	Yes — github.com/mullvad
Tagline	"Privacy is for the people" (homepage), "Privacy is a universal right" (GitHub)

A.5 Caveats and Warnings

Sweden is a 14 Eyes member. This is the single most-cited criticism of Mullvad. Sweden has agreed to share intelligence with the UK, US, Canada, Australia, New Zealand, and eight other countries. Mullvad's counter-argument: since there are no logs to hand over, the jurisdiction matters less. Court cases (2023 Swedish police case, 2024 German case) have validated that a non-deployment of logs is genuinely enforceable — Mullvad has consistently reported they have no logs to provide.

No free tier. Mullvad has no free tier (unlike ProtonVPN, Windscribe, etc.). The €5/month is the only option. Operating a privacy-respecting VPN costs money; Mullvad refuses to monetize via ads or data sales.

No 24/7 support. Support hours are Sweden business hours, weekdays only. Email support (PGP-encrypted for sensitive matters) is the only channel.

Bridges are NOT multihop. Bridges solve a different problem (censorship circumvention). SOCKS5 multihop (Chapter 18) is the privacy-focused exit-chaining feature. Don't confuse the two.

Part B — 90s Tutorial Walkthrough ("Now, you do it!")

B.1 What You'll Need

- A computer (macOS, Windows, or Linux)
- A working internet connection
- An email account (optional — for backup, not required)
- About 30 minutes

B.2 Objectives

In this walkthrough, you'll:

1. Visit Mullvad's homepage and explore the marketing site
2. Generate an account number
3. Add 30 days to your account
4. Verify your new IP

B.3 Step 1. Visit Mullvad's homepage.

Open a web browser and navigate to:

```
https://mullvad.net
```

You'll see Mullvad's homepage with their tagline "Privacy is for the people" prominently displayed. Take a moment to read the homepage hero copy. This is Mullvad's mission in their own words. It sets the tone for everything else you'll encounter.

Tip: For maximum privacy during signup, you can visit Mullvad's homepage via their Tor onion service instead. The onion URL is published on their homepage footer. Use Tor Browser to access it.

B.4 Step 2. Generate your account number.

Click the **"Generate account number"** button on the homepage. A 16-digit number will appear on screen. **Copy this number immediately** and save it somewhere safe — you won't be able to recover it later if you lose it.

Your account number looks like this (example, do not use):

```
1234 5678 9012 3456
```

Gotcha: You will NOT receive an email. You will NOT be asked for an email. This is by design. Without an email, Mullvad cannot accidentally leak your identity via a third-party email breach.

Important: Write this number down. Mullvad literally cannot recover it — they have no way to verify you are who you say you are. If you lose the number, you must create a new account (and start over with time on the account).

B.5 Step 3. Add time to your account.

Click **"Add time"** on the account page. Choose your payment method:

- **Credit card / PayPal / Swish** — easiest, fastest, but ties your identity to Mullvad

- **Cryptocurrency** (Monero or Bitcoin) — better privacy
- **Cash in envelope** — best privacy, slowest (mail to Sweden)
- **Voucher** — buy a code from an authorized reseller

For this walkthrough, use credit card to add 1 month (€5). In a follow-up chapter (Ch 36 — Pricing), we'll go deeper on anonymous payment.

B.6 Step 4. Visit the about page.

Open <https://mullvad.net/en/about/>. You'll see the full history of the company, including:

"March 2009 — The Mullvad VPN service launches!"

"Mullvad VPN AB is owned by parent company Amagicom AB. The name Amagicom is derived from the Sumerian word ama-gi — the oldest word for 'freedom' or, literally, 'back to mother' in the context of slavery — and the abbreviation for communication. Amagicom stands for 'free communication'."

"Mullvad VPN AB and its parent company Amagicom AB are 100% owned by founders Fredrik Strömberg and Daniel Berntsson who are actively involved in the company."

Tip: Mullvad's history page is genuinely interesting — it shows a 17-year journey from a small Gothenburg-based startup to one of the most-respected VPN providers globally.

B.7 Step 5. Skim the press page.

Open <https://mullvad.net/en/press/>. Note:

- The full mission statement (verbatim in Part A.2 above)
- The brand assets (logos, fonts, color palette)
- The official fonts: body = **Open Sans**, headlines = **Source Sans Pro**
- The official palette: dark blue #192E45, blue #294D73, green #44AD4D, red #E34039

This page is where journalists and authors (like the author of this book) get permission to use Mullvad's brand assets.

B.8 Summary

New concept	Meaning
Account number	16-digit number, no email or password — your credential
Mullvad VPN AB	Legal name (Gothenburg, Sweden)
Amagicom AB	Parent company, 100% owned by founders
14 Eyes	Intelligence-sharing alliance Sweden belongs to
RAM-only servers	No persistent storage; lose state on reboot
Cure53 / Assured AB	Independent audit firms

New command	Meaning
(no commands yet — that's for Chapter 7 on Linux CLI)	

B.9 Exercises

1. **Generate a throwaway account.** Use Mullvad's homepage to generate a test account number. Don't add time — just see the flow. → **What does the UI show you?**
2. **Try the onion service.** Open Tor Browser and navigate to the onion URL published on Mullvad's homepage footer. → **Does the site load? Does it look different from the clearnet version?**
3. **Read the privacy policy.** Open <https://mullvad.net/en/help/privacy-policy>. Count how many third-party trackers the site loads. → **Compare with another major VPN's privacy policy page. Who's cleaner?**

Part C — Platform-Specific Notes

C.1 macOS

- The Mullvad homepage renders well in Safari. Use **Get the app** → macOS for next steps (covered in Chapter 5).
- For maximum privacy during signup: use Safari with **Private Browsing** enabled (Cmd-Shift-N), or use Tor Browser.

C.2 Windows

- The Mullvad homepage works in Edge, Chrome, and Firefox identically.
- Use **InPrivate** (Edge) or **Incognito** (Chrome) for private browsing during signup.

C.3 Linux

- The Mullvad homepage works in all major browsers.
- For maximum privacy during signup: use Tor Browser (apt install torbrowser-launcher) and access via onion.

C.4 iOS / Android

- The Mullvad homepage is mobile-responsive. Works in Safari (iOS) and Chrome (Android).
- For maximum privacy: download Onion Browser (iOS) or Tor Browser for Android, and access via onion.

Part D — References

The resources below are referenced throughout this chapter. Each link has a QR code for instant scanning to your phone.

Resource	URL	QR
Mullvad homepage	https://mullvad.net	
About Mullvad	https://mullvad.net/en/about/	
Press page	https://mullvad.net/en/press/	
Privacy policy	https://mullvad.net/en/help/privacy-policy	
Terms of service	https://mullvad.net/en/help/terms-service	
Cookies policy	https://mullvad.net/en/help/cookie-policy	
All policies	https://mullvad.net/en/policies	
Mullvad on GitHub	https://github.com/mullvad	
Mullvad YouTube	https://www.youtube.com/c/MullvadVPNNet	
Mullvad on X	https://www.x.com/mullvadnet	
Mullvad on Mastodon	https://mastodon.online/@mullvadnet	

Closing Encouragement

That's the quick tour of Mullvad's purpose and philosophy. You now know:

- What Mullvad is (a Swedish VPN provider founded in 2009)
- Why they exist (to fight mass surveillance and censorship)
- Who owns it (the two founders, via Amagicom AB)
- What makes them different (account-number-only, anonymous payment, audited no-logs)
- How to start (generate an account, add time, log in)

In Chapter 2, we'll dig into *what VPNs can and cannot do* — the threat model. This is essential reading before you configure anything, because understanding what you're defending against shapes every decision you'll make about server choice, payment method, and feature enablement.

Have fun — and remember, privacy is for the people.

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- Google Play: https://play.google.com/books/GOOGLE_ID_PLACEHOLDER
- Samsung Books: https://www.draft2digital.com/book/D2D_ID_PLACEHOLDER
- Gumroad: <https://stephenlangley.gumroad.com/l/mullvad-full>
- Leanpub: <https://leanpub.com/mullvad-vpn>
- Smashwords: https://www.smashwords.com/books/view/SMASHWORDS_ID_PLACEHOLDER
- Lulu: <https://www.lulu.com/spot/stephenlangley/>
- itch.io: <https://stephenlangley.itch.io/mullvad-vpn>

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title: "The Threat Model" slug: "threat-model"

Chapter 2 — The Threat Model

Learning Objectives

By the end of this chapter, you'll understand:

- What a VPN can and cannot protect
 - The four-tier adversary categorization
 - How Mullvad's features map to each tier
 - When to use Mullvad, and when not to
-

Part A — Detailed Information

A.1 What a VPN Can Do

A VPN (Virtual Private Network) routes your traffic through a server operated by the VPN provider, replacing your real IP address with the VPN server's IP. This gives you several protections:

- **Hide your IP** from the websites you visit
- **Hide your DNS lookups** from your ISP (with proper DNS handling)
- **Hide the contents** of your traffic from your local network operator (your ISP, a hotel WiFi, etc.)
- **Bypass geographic restrictions** on content (Netflix, BBC iPlayer, etc.)
- **Bypass ISP-level censorship** of specific services

A.2 What a VPN Cannot Do

A VPN is not magic. It does NOT:

- **Make you anonymous** if you log into identifiable accounts (Google, Facebook, etc.)
- **Protect against malware, phishing, or social engineering**
- **Protect against browser fingerprinting** or cookies
- **Make you invisible** to a global passive adversary capable of timing attacks
- **Replace the need for good OpSec** (password managers, 2FA, etc.)

The fundamental law: *a VPN shifts your trust from your ISP to the VPN provider*. If the VPN provider logs your activity, you've gained nothing.

A.3 Adversary Categorization (for the book)

Tier 1 — Casual observers

Public WiFi operator, your ISP, a website you visit. **Mullvad fully protects against them.**

Tier 2 — Local law enforcement / civil litigants

Police in your country wanting to know "who connected to X", a copyright troll wanting to identify a downloader. **Mullvad fully protects** — there are no records to subpoena.

Tier 3 — National intelligence agency

NSA, GCHQ, FRA, etc. Capability: passive collection at IXPs, optional collaboration with ISP.

Mullvad partial protection. Multihop helps. End-to-end encrypted multihop (the WireGuard variant, Ch 19) provides the strongest protection. Your *metadata* (timing, volumes) may still be inferable.

Tier 4 — A Global passive adversary

A hypothetical adversary with access to multiple jurisdictions simultaneously.

Mullvad limited protection. Only multihop with both layers (WireGuard multihop + SOCKS5 third hop) makes this difficult. Still not impossible against a patient adversary.

A.4 When NOT to Use a VPN

- **Banking sites** often block VPN IPs (fraud prevention) — disable Mullvad to access
- **Streaming services with strong geo-restrictions** (Netflix, BBC iPlayer) may detect and block VPN IPs
- **When you need real identity** for a service (your bank's fraud detection)

A.5 When TO Use a VPN

- **Always** when on public WiFi
- **Always** when downloading or sharing content you don't want logged
- **Often** when you want geographic content distribution
- **Frequently** when working with sensitive information (journalism, activism)
- **As a baseline** for daily browsing to deny your ISP visibility

Part B — 90s Tutorial Walkthrough

B.1 What You'll Need

- A working browser
- An honest assessment of who you're worried about

B.2 Objectives

1. Map your concerns to a threat tier

2. Choose the right Mullvad features for that tier
3. Set up your account with appropriate payment

B.3 Step 1. Identify your adversary.

Write down — on paper, ideally — who you're worried about seeing your traffic. Common answers:

- "My ISP" → Tier 1
- "A nosy landlord or IT department" → Tier 1
- "Local law enforcement" → Tier 2
- "A foreign state-level adversary" → Tier 3
- "A global passive adversary" → Tier 4

B.4 Step 2. Pick the Mullvad features that match.

Tier	Recommended Mullvad features
1	Default config + ad-blocking DNS
2	Default config + Mullvad Browser + ad-blocking DNS
3	Multihop (WireGuard end-to-end) + DAITA + Monero payment
4	Multihop + DAITA + SOCKS5 third hop + Monero + onion service + dedicated laptop

B.5 Step 3. Choose your payment method.

Tier	Recommended payment
1	Credit card is fine
2	Credit card or voucher
3	Bitcoin via Tor or Monero
4	Cash in envelope + Monero

Gotcha: Payment method matters because **your payment ties to your identity** (bank, exchange, mailing envelope). A no-logs VPN can only protect you from the network layer; the payment layer is separate.

B.6 Summary

Tier	Adversary	Mullvad protection
1	Casual observers (ISP, public WiFi)	Complete
2	Local law / copyright trolls	Complete
3	National intelligence	Partial (multihop helps)
4	Global passive	Limited (only multi-layer helps)

B.7 Exercises

1. **Write down your threat model.** Spend 10 minutes listing: who are you protecting from? What are you protecting? What trade-offs are you willing to make? → Keep this as a reference.
2. **Match your threat to Mullvad features.** Use the table in Step 2 to pick your configuration. → Be honest about whether you're Tier 1 or Tier 3.
3. **Check your browser fingerprint.** Visit <https://coveryourtracks.eff.org> (run by the EFF) → See how unique your browser fingerprint is. Mullvad Browser masks this; regular Chrome/Firefox do not.

Part C — Platform-Specific Notes

The threat model applies uniformly across platforms. What's different is the *implementation* — Mullvad Browser, kill switch settings, etc.

Platform	Best practice
macOS	Enable kill switch; use Mullvad Browser for sensitive browsing
Windows	Enable "Always require VPN"; disable IPv6
Linux	Configure iptables-based kill switch; mullvad status for monitoring
iOS	Always-on VPN; Mullvad Browser for sensitive browsing
Android	Always-on VPN + Block connections without VPN

Part D — References

Resource	URL	QR
Mullvad homepage	https://mullvad.net	
Mullvad on X	https://www.x.com/mullvadnet	
EFF Cover Your Tracks	https://coveryourtracks.eff.org	

Closing Encouragement

The threat model is the foundation of every privacy decision you'll make. If you skipped it because it felt abstract — go back and write yours down. **Have fun — and think before you click.**

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- Samsung Books: https://www.draft2digital.com/book/D2D_ID_PLACEHOLDER
- Gumroad: <https://stephenlangley.gumroad.com/l/mullvad-full>
- Leanpub: <https://leanpub.com/mullvad-vpn>
- Smashwords: https://www.smashwords.com/books/view/SMASHWORDS_ID_PLACEHOLDER
- Lulu: <https://www.lulu.com/spot/stephenlangley/>
- itch.io: <https://stephenlangley.itch.io/mullvad-vpn>

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title: "Multihop with SOCKS5" slug: "multihop-with-socks5"

Chapter 18 — Multihop with SOCKS5

This is the showcase chapter. You'll learn the architecture that powers SOCKS5 multihop on Mullvad, then walk through a hands-on tutorial that connects you to a Swedish server while your browser exits through the Netherlands.

Learning Objectives

By the end of this chapter, you'll be able to:

- Explain in your own words what Multihop with SOCKS5 is and why it exists
- Configure Firefox to exit your traffic through a different country than your WireGuard tunnel
- Verify your exit IP and DNS are different from your entry tunnel
- Understand the security trade-offs of SOCKS5 multihop vs. WireGuard end-to-end multihop

Part A — Detailed Information

A.1 What Multihop with SOCKS5 Is

Multihop with SOCKS5 is the privacy feature that lets your browser traffic **exit** through a different country than the WireGuard tunnel you connected through. Concretely:

"By connecting to any of our WireGuard® servers and configuring your browser (or other SOCKS5 compatible software) to use another WireGuard server's SOCKS5 proxy, the browser's traffic will EXIT in a different location than the WireGuard server you are connecting to. For example, you can connect using the Mullvad VPN app to a WireGuard server in Sweden, and configure your browser to use a SOCKS5 proxy in the US. Your browser traffic will then first enter a server in Sweden and then exit through a server the US."

— Mullvad Help: <https://mullvad.net/en/help/different-entryexit-node-using-wireguard-and-socks5-proxy>

In other words:

Without SOCKS5 multihop:

```

You → [WireGuard tunnel] → Exit Server → Internet
      ↑ entry ↓ from your ISP ↑

```

With SNRKS5 multihop.

```

You → [WireGuard tunnel] → Entry Server → SOCKS5 (different server) → Internet
      ↑ entry ↓ from your ISP      ↑ exit ↑ from different country ↑

```

The wireguard tunnel protects your traffic as it leaves your network. The SOCKS5 proxy at a different exit server determines where your traffic **appears to come from** to the websites you visit.

A.2 Why It Matters (Threat Model Fit)

SOCKS5 multihop helps against:

- **Timing correlation attacks** — splitting your entry and exit across jurisdictions makes it harder for an adversary to correlate your entry traffic with your exit traffic
- **ISP-level logging** — your ISP only sees encrypted WireGuard traffic; the destination websites only see the exit server's IP
- **Cloudflare CAPTCHAs** — sites that block known VPN IPs will still see the exit IP; SOCKS5 multihop with the proxy's stable IP reduces CAPTCHAs (per Mullvad's docs)
- **IP-based whitelisting** — the SOCKS5 proxy provides a stable IP for whitelisting (the server's IP, not the dynamic exit of a regular tunnel)

It does **NOT** help against:

- **Root access to the entry server** — see the caveat in A.5 below
- **Browser fingerprinting** — use Mullvad Browser to address that

A.4 Architecture (The Two SOCKS5 Endpoints Per Server)

Mullvad's WireGuard servers expose exactly **two SOCKS5 proxies**:

SOCKS5 endpoint	Address	Reachable from where?	Purpose
Local	10.64.0.1:1080	Inside the WireGuard tunnel to this same server	The "single-hop" SOCKS5; same exit IP as the connected server
Peer-reachable	10.124.0.x:1080	From any other WireGuard server, via Mullvad's inter-server overlay	SOCKS5 multihop exit; .x is unique per server

Verbatim from Mullvad:

"All our WireGuard servers have two SOCKS5 proxies listening on them: • The SOCKS5 proxy on 10.64.0.1 with port 1080 which is not reachable from other WireGuard servers. • The SOCKS5 proxy on 10.124.0.x to 10.124.1.x with port 1080 which is reachable from other WireGuard servers. These IPs are unique for each WireGuard server. For instance, 10.124.0.4 belongs to nl-ams-wg-001 and 10.124.0.2 belongs to se-mma-wg-001."

— Mullvad Help: <https://mullvad.net/en/help/socks5-proxy/> (dateModified 2026-07-30)

For browser-based SOCKS5 multihop, you use the **public hostname** that DNS-resolves to one of Mullvad's servers. The hostname is:

```
xx-xyz-wg-socks5-###.relays.mullvad.net
```

For example, to exit via `nl-ams-wg-001`, you use `nl-ams-wg-socks5-001.relays.mullvad.net:1080`.

A.5 Specifications (Verified Live, 2026-09-28)

Property	Value
SOCKS5 endpoints	Two per WireGuard server
SOCKS5 port	1080/tcp (always)
Local address	10.64.0.1 (always the same; per-server routing handles the magic)
Peer address	10.124.0.x through 10.124.1.x (unique per server)
SOCKS5 protocol version	SOCKS5
Encryption	None at SOCKS5 level; HTTPS on top protects content
DNS handling	"Proxy DNS when using SOCKS v5" recommended (browser-level)
Total WireGuard servers	553
Verified example	se-mma-wg-001 → SOCKS5 alias se-mma-wg-socks5-001.relays.mullvad.net
Verified example	nl-ams-wg-001 → SOCKS5 alias nl-ams-wg-socks5-001.relays.mullvad.net

A.6 Caveats and Warnings (Verbatim, Reproduced Prominently)

▲ THE BIG SECURITY CAVEAT — PLEASE READ BEFORE USING

"Note: This technique to jump from one server to another is generally called Multihop, however, even though all traffic between the WireGuard servers are encrypted, if someone has root access to the first server and the user data transported is not encrypted, the person with root access can see the data. If the data is encrypted with HTTPS the person with root access can see the domain names that are accessed (root access is something that only sysadmins at Mullvad have) and IP-addresses used. This is no different from our normal OpenVPN solution. However Multihop using our WireGuard guide protects even from root access to the entry server since it has end-to-end encryption. If you use the WireGuard configuration generator to enable multihop you can still use SOCKS5 as described here and add a third hop."

In plain English: **SOCKS5 multihop does not protect against root access to the entry server.** If you're worried about that, use **WireGuard multihop** (Chapter 19) for end-to-end encryption.

Other caveats:

- The SOCKS5 proxy only works when you're connected to Mullvad VPN. If you disconnect, the proxy is unreachable. (This is actually a feature: it's a kill switch for the browser.)

- "Proxy DNS when using SOCKS v5" must be enabled in Firefox for DNS leak protection. Enabling it disables the Mullvad app's DNS content blockers.
 - Edge (Windows) does NOT enable IPv6 SOCKS5 proxy — only IPv4.
-

Part B — 90s Tutorial Walkthrough ("Now, you do it!")

B.1 What You'll Need

- A working Mullvad VPN subscription (Ch 1)
- The Mullvad app installed and connected to `se-mma-wg-001` (or any server; we'll use Sweden for this walkthrough)
- Firefox installed (we'll use Firefox for this walkthrough)
- A terminal (Terminal.app on macOS, PowerShell on Windows, gnome-terminal / xterm on Linux)
- About 20 minutes

B.2 Objectives

In this walkthrough, you'll:

1. Confirm your baseline (no SOCKS5)
2. Configure Firefox with `10.64.0.1:1080` (single-hop SOCKS5)
3. Configure Firefox with `nl-ams-wg-socks5-001.relays.mullvad.net:1080` (multi-hop SOCKS5)
4. Verify each step's exit IP

B.3 Step 1. Confirm your baseline (no SOCKS5).

Open your terminal and run:

```
curl https://am.i.mullvad.net
```

The output is your current exit IP (the IP of the Mullvad server you're connected to). In our walkthrough, you'll see a Swedish IP — because we connected to `se-mma-wg-001`. Write it down.

Tip: The `am.i.mullvad.net` endpoint exists specifically for this purpose. There's an IPv4 variant (`ipv4.am.i.mullvad.net`) and an IPv6 variant (`ipv6.am.i.mullvad.net`) for testing each protocol.

B.4 Step 2. Connect to `se-mma-wg-001`.

Open the Mullvad app. **Switch location** → **Sweden** → `se-mma-wg-001`. Wait for the connection to establish.

Alternatively via the Mullvad CLI:

```
mullvad relay set hostname se-mma-wg-001  
mullvad connect
```

Confirm by re-running:

```
curl https://am.i.mullvad.net
```

You should still see a Swedish IP (since you're connected to a Swedish server).

B.5 Step 3. Configure Firefox to use the single-hop SOCKS5 proxy.

Open Firefox. Navigate to:

```
about:preferences#general
```

Scroll down to **Network Settings** → click **Settings...** The Connection Settings dialog opens.

A popup options box:

```
Configure Proxy Access:
o Use system proxy settings
o Auto-detect proxy settings
o Use automatic proxy configuration script
● Manual proxy configuration

SOCKS Host: 10.64.0.1
SOCKS Port: 1080

[✓] Proxy DNS when using SOCKS v5
```

Click the radio button for **Manual proxy configuration**. In the **SOCKS Host** field, type `10.64.0.1`. In the **SOCKS Port** field, type `1080`. Tick the checkbox **"Proxy DNS when using SOCKS v5"** (this is critical for DNS leak protection). Click **OK**.

Gotcha: Firefox has separate fields for HTTP Proxy and SOCKS Host. Make sure you're filling in the SOCKS fields, not HTTP. The HTTP fields should be **empty**.

Tip: "Proxy DNS when using SOCKS v5" routes DNS through the SOCKS5 tunnel. Without this, your browser would resolve DNS via your system DNS (likely leaking to your ISP), even though the connection itself is tunneled.

B.6 Step 4. Verify the single-hop SOCKS5 proxy.

Open a **new tab** in Firefox (the existing tab may have cached the old connection). Visit:

```
https://am.i.mullvad.net
```

You should see a **different Swedish IP** than your baseline. That's because the SOCKS5 proxy has its own stable exit IP (different from the WireGuard tunnel's dynamic exit IP).

Tip: Open a **private window** (Cmd-Shift-P on macOS, Ctrl-Shift-P on Windows/Linux) to avoid cookies and cache influencing the test. Private windows start fresh.

Confirm by visiting the same URL in your terminal:

```
curl https://ipv4.am.i.mullvad.net --socks5-hostname 10.64.0.1
```

This is the command-line equivalent. You should see the same IP as Firefox shows.

B.7 Step 5. Reconfigure Firefox for multi-hop SOCKS5 (the real Multihop).

Now the magic step. Go back to **about:preferences#general** → **Network Settings** → **Settings....**

Replace:

```
SOCKS Host: nl-ams-wg-socks5-001.relays.mullvad.net
SOCKS Port: 1080
```

The SOCKS Host is now a hostname, not an IP. Click **OK**.

Gotcha: Make sure you type the hostname EXACTLY. The `relays.mullvad.net` suffix is required. A typo will give you a confusing error.

B.8 Step 6. Verify multi-hop SOCKS5 (the showcase).

Open a **new tab** (and/or private window). Visit:

```
https://am.i.mullvad.net
```

You should now see a **Dutch IP** — even though your WireGuard tunnel is still going to Sweden. That's the multihop in action: Swedish entry, Dutch exit.

Verify with the command line:

```
curl https://ipv4.am.i.mullvad.net --socks5-hostname nl-ams-wg-socks5-001.relays.mullvad.net
```

Same Dutch IP.

Tip: The curl command should be from the SAME machine — not from inside Firefox. Your terminal will use your system's SOCKS5 settings (or the explicit `--socks5-hostname` flag). Firefox uses its own SOCKS5 settings.

B.9 Step 7. Confirm IPv6 exit.

Mullvad's SOCKS5 proxies support dual-stack. To verify your IPv6 exit:

```
curl https://ipv6.am.i.mullvad.net --socks5-hostname nl-ams-wg-socks5-001.relays.mullvad.net
```

You'll see a `2a03:1b20:...` style IPv6 address (Mullvad uses this AS range). The exit IP for IPv6 will be different from your IPv4 exit — that's expected (each proxy has both an IPv4 and IPv6 address, both unique).

B.10 Step 8. Confirm DNS is not leaking.

Visit <https://dnsleaktest.com> and click **Extended Test**. After the test runs, the DNS servers shown should NOT be your ISP's. They should be either Mullvad's DNS (set by the tunnel) or "no servers found" (your browser's DNS is going through the SOCKS5 tunnel, which is the intended behavior).

If you see your ISP's DNS, double-check:

1. Firefox's "Proxy DNS when using SOCKS v5" is ticked
2. The SOCKS Host is `nl-ams-wg-socks5-001.relays.mullvad.net` (or another valid SOCKS5 alias), not an IP that doesn't resolve
3. Your Mullvad tunnel is still up (the SOCKS5 proxy only works inside the tunnel)

B.11 Step 9. Restore your normal settings.

When you're done testing, restore Firefox's network settings to **Use system proxy settings** (or whatever you had before). Otherwise all your browser traffic will keep going through the Dutch exit.

You can also disable the Mullvad tunnel (Settings → Disconnect) when you're done — Firefox will then have no proxy to use and won't be able to reach the internet.

Gotcha: This is actually the kill-switch bonus of SOCKS5 multihop. If you forget to disable the SOCKS5 proxy in Firefox and you disable the Mullvad tunnel, Firefox will refuse to connect to anything — preventing accidental leaks.

B.12 Summary

New concept	Meaning
Multihop with SOCKS5	WireGuard tunnel + SOCKS5 proxy at a different server
<code>10.64.0.1:1080</code>	The single-hop SOCKS5 (same exit as your tunnel)
<code>xx-xyz-wg-socks5-###.relays.mullvad.net</code>	The hostname for a specific server's SOCKS5
Multi-hop SOCKS5	Tunnel to one server, exit through a different one
DNS leak	Without "Proxy DNS when using SOCKS v5", DNS may leak
Kill-switch bonus	If the tunnel drops, the SOCKS5 proxy becomes unreachable

New command	Meaning
<code>curl https://am.i.mullvad.net</code>	See your current exit IP
<code>curl --socks5-hostname 10.64.0.1 <url></code>	Route through the single-hop SOCKS5
<code>curl --socks5-hostname nl-ams-wg-socks5-001.relays.mullvad.net <url></code>	Route through the multihop SOCKS5
<code>mullvad relay set hostname se-mma-wg-001</code>	Pick a specific WireGuard tunnel entry

B.13 Exercises

1. **Try a different exit country.** Pick a different SOCKS5 alias — e.g., `us-nyc-wg-socks5-204.relays.mullvad.net` (if it exists in the API). Set it in Firefox. Visit `am.i.mullvad.net` . → **What country does your exit IP show?**
2. **Test DNS leak prevention.** With multihop configured, visit `https://dnsleaktest.com`. Run the extended test. → **What DNS servers does it report?**
3. **Test IPv6 + IPv4 simultaneously.** Run both curl commands in two terminal tabs:

```
curl https://ipv4.am.i.mullvad.net --socks5-hostname nl-ams-wg-socks5-001.relays.mullvad.net curl https://ipv6.am.i.mullvad.net --socks5-hostname nl-ams-wg-socks5-001.relays.mullvad.net
```

→ **Are the IPs in the same country? Are they from the same subnet?**

1. **Test the kill switch.** With Firefox configured for SOCKS5 multihop, disconnect the Mullvad app. Try to load any page in Firefox. → **What happens?** (Hint: it should refuse to connect — that's the kill switch.)

Part C — Platform-Specific Notes

C.1 macOS / Windows / Linux (Firefox)

Same steps as above. Firefox's network settings dialog is identical across platforms.

C.2 Chrome / Chromium (Command-Line Flag)

Chrome doesn't expose SOCKS5 settings in the GUI — you need to launch it from the command line:

```
chromium-browser --proxy-server=socks5://10.64.0.1:1080
```

For multi-hop:

```
chromium-browser --proxy-server=socks5://nl-ams-wg-socks5-001.relays.mullvad.net:1080
```

To force WebRTC to not leak:

```
chromium-browser \
  --proxy-server=socks5://nl-ams-wg-socks5-001.relays.mullvad.net:1080 \
  --force-webrtc-in-handling-policy \
  --webrtc-ip-handling-policy=disable_non_proxied_udp
```

C.3 Firefox About:Config DNS

By default, Firefox uses the system DNS. The "Proxy DNS when using SOCKS v5" option in the network settings dialog overrides this for SOCKS5 connections.

For tighter control, visit `about:config` and set: - `network.proxy.socks` = `nl-ams-wg-socks5-001.relays.mullvad.net` - `network.proxy.socks_port` = `1080` - `network.proxy.socks_remote_dns` = `true` - `network.proxy.type` = `1` (manual)

C.4 Edge (Windows)

Edge's SOCKS5 support is similar to Chrome, but: **Edge does NOT enable IPv6 SOCKS5 proxy** (only IPv4). If you need IPv6 exit, use Firefox or Chrome.

C.5 Mullvad Browser

Mullvad Browser includes a Mullvad extension that lets you switch location easily. Click the Mullvad icon in the toolbar → pick a city → done. No command-line flags, no `about:config`.

C.6 Per-App Configs

Many apps support SOCKS5 proxies:

App	How to configure SOCKS5
Firefox	Settings → Network → Manual proxy
Chrome / Edge	<code>--proxy-server=socks5://...</code> flag
Telegram	Settings → Advanced → Connection type → SOCKS5
Thunderbird	Settings → General → Network → Manual proxy
curl	<code>--socks5-hostname <host></code>
SSH	<code>ssh -D 1080 ...</code> then configure app to use <code>127.0.0.1:1080</code>

For apps that don't support SOCKS5, use `tsocks` or `proxychains` on Linux to force the app's traffic through the SOCKS5 proxy.

Part D — References

The resources below are referenced throughout this chapter. Each link has a QR code for instant scanning.

Resource	URL	QR
SOCKS5 proxy guide (Mullvad)	https://mullvad.net/en/help/socks5-proxy/	[QR:socks5_proxy_doc]
Multihop with SOCKS5 + WG (Mullvad)	https://mullvad.net/en/help/different-entryexit-node-using-wireguard-and-socks5-proxy	[QR:multihop_socks5_wg_doc]
Am I Mullvad? connection check	https://am.i.mullvad.net	[QR:am_i_mullvad]
IPv4 connection check	https://ipv4.am.i.mullvad.net	[QR:ipv4_am_i_mullvad]
IPv6 connection check	https://ipv6.am.i.mullvad.net	[QR:ipv6_am_i_mullvad]
Mullvad homepage	https://mullvad.net	
Mullvad WireGuard relays API	https://api.mullvad.net/www/relays/wireguard/	[QR:api_relays_wireguard]
Mullvad on GitHub	https://github.com/mullvad	[QR:github_mullvad]
DNS leak test	https://dnsleaktest.com	(see Ch 14 for full reference)

Closing Encouragement

You just walked through the architecture that powers one of the most distinctive Mullvad features. You can now:

- Explain why there are two SOCKS5 endpoints per server

- Configure Firefox (and Chrome/Edge via flags) for both single-hop and multi-hop
- Verify your exit IP, IPv6 address, and DNS leak status
- Know that SOCKS5 multihop has caveats (no entry-server protection) and when to use WireGuard multihop instead

In **Chapter 19 (Multihop with WireGuard)**, we'll explore the end-to-end encrypted variant — which protects against root access to the entry server. **Have fun — and remember, the only person who knows your traffic is you.**

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- Gumroad: <https://stephenlangley.gumroad.com/l/mullvad-full>
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- Smashwords: https://www.smashwords.com/books/view/SMASHWORDS_ID_PLACEHOLDER
- Lulu: <https://www.lulu.com/spot/stephenlangley/>
- itch.io: <https://stephenlangley.itch.io/mullvad-vpn>

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title: "Glossary" slug: "glossary"

Chapter 38 — Glossary

A reference of VPN + Mullvad-specific terms used throughout this book.

VPN + Tunneling

Bridge (server) — An OpenVPN + Shadowsocks endpoint designed to bypass censorship. Has no SOCKS5 capability. Naming: `xx-xyz-br-###.relays.mullvad.net`.

Bridge mode — A Mullvad app setting that routes your traffic through a Shadowsocks bridge server. For restrictive networks.

DAITA — **D**efense **A**gainst **I**ntegral **a**ssistants of **T**raffic **A**nalysis. Mullvad's patented traffic-analysis resistance feature. Server capability flag: `daita: true/false`.

DNS leak — When DNS queries leak outside the VPN tunnel, exposing the websites you visit to your ISP. Mullvad's kill switch + custom DNS prevent this.

IP leak — When your real IP address is exposed despite being connected to the VPN. Usually due to WebRTC (browser), IPv6 leak, or app misconfiguration.

Kill switch — A setting that blocks all network traffic if the VPN tunnel drops. Prevents accidental unencrypted traffic.

Multihop — Routing your traffic through two (or more) VPN servers in sequence. Privacy benefit against timing-correlation attacks.

Shadowsocks — A SOCKS5-based proxy protocol designed to look like normal HTTPS traffic. Used by Mullvad's Bridge mode.

Split tunneling — Routing some apps through the VPN while others use the direct internet connection. Power-user feature.

WireGuard — A modern VPN protocol. Fast, simple, audited. Mullvad's preferred protocol (since ~2020).

OpenVPN — An older but still popular VPN protocol. Slower than WireGuard but more flexible. Mullvad supports it for Bridge mode.

Mullvad-Specific

Account number — A 16-digit number that serves as your only login credential. No email, no password, no name.

Amagicom — The parent company of Mullvad VPN AB. 100% owned by founders Fredrik Strömberg and Daniel Berntsson.

Cure53 — Berlin security firm that audited Mullvad in 2020.

Assured AB — Swedish security firm that audited Mullvad's no-logs policy in 2023.

DAITA server — A WireGuard relay with `daita: true` in the API. Required for DAITA-enabled connections.

Direct only — The legacy setting name for "when a feature requires multihop, use direct connection" (replaced by "When needed" mode in 2026).

Entry server — The WireGuard server you connect to first. The "input" of your tunnel.

Exit server — The server your traffic exits to the destination website. Where your traffic "appears to come from" to the outside world.

Mullvad Browser — A Mullvad-curated Firefox fork with privacy hardening. Free, separate from the VPN app.

Mullvad VPN AB — The legal entity operating Mullvad VPN. Swedish reg. no. 559238-4001.

Multihop_port — The per-server WireGuard UDP listen port. Range 3002–3622 (verified 2026-09-28). Misleading name; it's the tunnel endpoint, not the SOCKS5 port.

SOCKS5 alias — The DNS hostname for a server's SOCKS5 listener: `xx-xyz-wg-socks5-###.relays.mullvad.net`.

Stboot — Mullvad's "secure boot" server property. Servers with `stboot: true` boot from a verified image.

10.64.0.1 — The single, fixed SOCKS5 listener address on every WireGuard server. Reachable only from inside the tunnel to that server.

10.124.0.0/16 — Mullvad's inter-server overlay network. Each WG server has a unique IP in this range.

Security + Privacy

Anonymity — Not being identifiable. Distinct from privacy (which is "not being observed").

Jurisdiction — The country whose laws apply to a service. Sweden is Mullvad's jurisdiction.

14 Eyes — An intelligence-sharing alliance of 14 countries (UK, US, Canada, AU, NZ + 9 others). Sweden is a 14-Eyes member.

Threat model — A structured analysis of who you're protecting against, what you're protecting, and what trade-offs you're willing to make.

Warrant canary — A signed statement that the service has NOT received a secret gag order. If the canary disappears, infer a gag order.

Wintun — Windows tunneling driver used by WireGuard on Windows.

Network

Endpoint — In WireGuard config, the `Endpoint = host:port` line specifies where the tunnel terminates.

IXP — Internet Exchange Point. Where ISPs interconnect.

MTU — Maximum Transmission Unit. The largest packet size. Mullvad recommends 1380 for WireGuard over most networks.

NAT — Network Address Translation. Hides multiple private IPs behind one public IP.

TCP — Transmission Control Protocol. Reliable, ordered, slower.

UDP — User Datagram Protocol. Unreliable, faster, used by WireGuard.

WireGuard protocol — UDP-based VPN. Fast, modern, in-kernel on Linux.

Payment

Monero (XMR) — A cryptocurrency with strong privacy properties. Mullvad accepts it.

Bitcoin (BTC) — The first cryptocurrency. Less private than Monero but Mullvad accepts it.

Voucher — A pre-paid code redeemable for Mullvad time. Sold by resellers.

Cash in envelope — Most anonymous payment method. Send Swedish kronor to Mullvad's address.

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- Samsung Books: https://www.draft2digital.com/book/D2D_ID_PLACEHOLDER
- Gumroad: <https://stephenlangley.gumroad.com/l/mullvad-full>
- Leanpub: <https://leanpub.com/mullvad-vpn>
- Smashwords: https://www.smashwords.com/books/view/SMASHWORDS_ID_PLACEHOLDER
- Lulu: <https://www.lulu.com/spot/stephenlangley/>
- itch.io: <https://stephenlangley.itch.io/mullvad-vpn>

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title: "Frequently Asked Questions" slug: "faq"

Chapter 39 — Frequently Asked Questions

Compiled from the official Mullvad Help FAQ plus community questions.

General

Q: Why €5 flat? Why not a free tier?

Mullvad has no free tier. The flat €5 is intentionally simple — no upsells, no "premium" tier. Operating a privacy-respecting VPN costs money (servers, bandwidth, audits, staff); €5 covers it without surveillance-based monetization.

Q: Is Mullvad based in a 14-Eyes country?

Yes — Sweden is a 14-Eyes member. However, Sweden has strong domestic privacy laws (Personal Data Act + GDPR). Mullvad argues (and many privacy advocates agree) that **what matters is the no-logging policy**, not the jurisdiction, because there's nothing to hand over even if compelled.

Q: Has Mullvad ever handed over user data?

Mullvad publishes transparency reports. In every case where police or courts requested logs of a specific account, Mullvad had no data to provide (because of the no-logging policy + RAM-only infrastructure).

Q: Do you support port forwarding?

Yes. Mullvad supports port forwarding since 2024. Configure via the Mullvad app or CLI.

Q: Can I use Mullvad for torrenting / P2P?

Yes. Mullvad allows P2P traffic on all servers. Use the WireGuard protocol for best speeds.

Account + Payment

Q: I lost my account number. Can you recover it?

No. Mullvad cannot recover your account number — they don't store any identifying info. Create a new account.

Q: Can I pay with crypto?

Yes — Bitcoin and Monero. Mullvad also accepts cash in envelope and credit cards.

Q: Can I pay anonymously?

Yes — cash in envelope (most anonymous), Monero, or Bitcoin via Tor.

Q: Do you accept PayPal?

PayPal support was deprecated in ~2024. Use crypto, cash, or credit card instead.

Technical

Q: WireGuard or OpenVPN?

WireGuard is faster, simpler, and Mullvad's recommended default. Use **OpenVPN** only for Bridge mode (censorship circumvention).

Q: How do I multihop?

- **Easiest:** enable Multihop in the Mullvad app (Settings → Advanced → Multihop)
- **More control:** use SOCKS5 multihop (configure your browser to use `nl-ams-wg-socks5-001.relays.mullvad.net:1080`)

Q: Why do I get CAPTCHAs all the time?

"This is a known issue that happens on many sites using Cloudflare. Please follow our SOCKS5 guide to reduce the occurrence of CAPTCHAs, or you can try switching to another server." — Mullvad FAQ id:8

Q: Is Shadowsocks available in the app?

"Yes. By using Bridge mode you will be using the Shadowsocks proxy." — Mullvad FAQ id:54

Q: How do I get around censorship?

"Activate Bridge mode in the app. This will make use of the Shadowsocks proxy." — Mullvad FAQ id:55

Q: How fast is Mullvad?

20 Gbps server ports (verified across samples). Real-world throughput depends on your ISP, the server's distance, and current load. Typically 500 Mbps - 2 Gbps for nearby servers.

Q: Can I use Mullvad on my router?

Yes. See the router deployment chapter for OpenWrt and pfSense guides.

Q: Can I use Mullvad with Netflix / BBC iPlayer?

Streaming services often detect and block VPN IPs. Mullvad tries to keep working with major services, but there's no guarantee. Switch servers if one gets blocked.

Q: How many devices can I use simultaneously?

5 devices per account, enforced server-side.

Privacy

Q: Do you keep logs of my activity?

No. See the no-logging policy chapter for the full policy + audit history.

Q: Do you keep logs of my IP address?

No — your real IP is only seen during WireGuard handshake, then discarded. RAM-only servers mean it's gone on reboot anyway.

Q: Can I use Mullvad with Tor?

Yes. Connect to Mullvad, then open Tor Browser. The Tor exit node sees traffic from a Mullvad IP, not your real IP.

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- Samsung Books: https://www.draft2digital.com/book/D2D_ID_PLACEHOLDER
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